

HANIF EDMA FAUZ

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EDUCATION

Seoul National University of Science and Technology (SEOULTECH)

Master of Science, 4.19/4.50

Seoul, South Korea

Feb 2024 – Aug 2026

- Research assistant scholarship
- Major in electrical and information engineering, minor in control and robotic engineering
- Thesis: “Risk-Aware MPPI Control Using Gaussian Processes for Enhancing Mobile Robot Navigation Performance”

Universitas Indonesia

Bachelor of Electrical Engineering, 3.61/4.00

Depok, Indonesia

Aug 2019 – Aug 2023

- Major in electrical engineering, minor in control engineering
- Thesis: “Design of Adaptive Extended Kalman Filter to Estimate State of Charge of Lithium-Ion Battery”

EXPERIENCE

Control & Intelligent Robotics Lab SEOULTECH

Research Assistant

Seoul, South Korea

Feb 2024 – Aug 2026

- Developed a novel control framework and published findings
- Assisted professor with various academic activities

Train4best Indonesia

Trainer

Surabaya, Indonesia

Sep 2023

- Invited as a trainer to deliver an 8-day IoT platforms program at Politeknik Elektronika Negeri Surabaya (PENS)
- Delivered comprehensive hands-on training on IoT platforms to 10 vocational lecturers

DoctorTool (PT Medifa Infoyasa Suryantara)

Engineer Intern

Jakarta, Indonesia

Feb 2022 – Jul 2022

- Developed an ESP32-based IoT gateway bridging multiple medical instruments (thermometers, oximeters, biometric sensors) to the cloud, supporting over six device categories

PUBLICATIONS

- “Performance Enhanced Risk-Aware MPPI using Gaussian Process for Robust Mobile Robot Control”, *IEEE Access*, 2025, **1st Author**
- “Robust RA-MPPI using Online Learning”, *IJCAS*, Springer, **[Submitted], 1st Author**

OTHER PROJECTS

Web & Android Apps

Self-initiated side projects

hanifedma.com/portfolio

2023 – 2026

- Corrosion Detection: Flutter computer-vision app for Android, Windows, and Linux; 1st place at SciHackfest 2023 (PT Sucofindo), 25 million IDR (~1,600 USD) prize, team of 3
- Five cross-platform web and Android apps (vanilla JS; Kotlin / Jetpack Compose; Firebase / Supabase): CanopyDiary, Ponder, Waterline, TimberTimer, and Streak
- WhiteRose: Wear OS app – minute-interval wrist alerts (Kotlin)

SKILLS & LANGUAGES

- **Skills:** Electrical engineering, control engineering (MPC, PID, LQR, Kalman Filter), IT, robotics engineering, machine learning (Gaussian process, neural network, imitation learning)
- **Software:** C/C++, ROS1/ROS2, Gazebo, Linux, Git, Docker, MATLAB, Python, TensorFlow and PyTorch (basic), Web Development (basic), Flutter (basic)
- **Hardware:** Arduino, ESP32, Raspberry Pi, NVIDIA Jetson
- **Languages:** Bahasa Indonesia (native), English (C1, IELTS Academic, Band 7, 2023)